

Notes: Freed & Hackney (JFE, 2026)

Financial Literacy and Financial Crime: A Regression Discontinuity Approach

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Virginia financial literacy requirement	2
2.2	Administrative criminal data	2
2.3	Financial-crime definition	3
2.4	Control variables and fixed effects	3
2.5	Mechanism datasets	3
3	Models and empirical specifications	4
3.1	Sharp regression discontinuity design	4
3.2	Baseline outcome	4
3.3	Component-crime outcomes	4
3.4	Severity outcomes	4
3.5	Income heterogeneity	5
3.6	Indigent-defense outcome	5
3.7	Financial behavior outcomes	5
4	Identification and empirical design	5
4.1	What identifies the effect	5
4.2	Why the design is credible	6
4.3	Why this is not just a general education effect	6
4.4	Alternative mechanisms considered	6
5	Tables and figures	6
5.1	Figure 1 and Table 1: data granularity and summary statistics	6
5.2	Figure 2, Figure 3, and Table 2: first stage and baseline RDD	7
5.3	Table 3: robustness	8
5.4	Figure 4 and Table 4: placebo thresholds and crime types	9
5.5	Figure 5 and Figure 6: financial-crime composition and type-specific RDDs	10
5.6	Table 5 and Table 6: severity and non-financial crime	11
5.7	Figure 7 and Table 7: income and financial constraints	11
5.8	Table 8, Figure 8, and Table 9: financial distress and financial behavior	12
6	Short summary	13
7	Conclusion	14
7.1	Core conclusion	14
7.2	Economic intuition	14
7.3	Takeaway	14

1 Big picture

- **Question.** Does financial literacy reduce the propensity of individuals to commit financial crime?
- **Setting.** The paper studies a Virginia high-school policy requiring a stand-alone financial literacy course for students in the 2015 graduating cohort and later.
- **Identification.** The policy creates a sharp cutoff at the September 30, 1996 birth-date threshold. Individuals born just after the cutoff are assigned to the treated cohort; individuals born just before are assigned to the untreated cohort.
- **Data innovation.** The authors link administrative Virginia court records on criminal charges to residential addresses and demographic information for a near-complete sample of Virginia residents.
- **Main sample.** The full data contain about 5.5 million adults. The main RDD sample focuses on individuals ages 18–24 and, under the preferred bandwidth and fixed effects, contains 118,689 individuals.
- **Main result.** Exposure to high-school financial literacy reduces the probability of being charged with financial crime by about 32–37% relative to the sample mean.
- **Crime type.** The effect is concentrated in embezzlement, the financial crime most closely connected to financial stress and non-shareable financial problems.
- **Mechanism.** The strongest evidence points to a relaxation of financial constraints: treated individuals have lower high-interest credit-card debt, higher investment balances, and higher savings-account ownership.
- **Heterogeneity.** The effect is stronger for individuals from low-income Census block groups.
- **Alternative explanations.** The paper tests and finds little support for criminal sophistication, differential detection, out-migration, victimhood effects, school selection, or risk-preference channels.
- **Economic takeaway.** Financial education can generate social benefits beyond improved household finance: it can reduce financial crimes committed by ordinary individuals facing financial pressure.

2 Data and measurement

2.1 Virginia financial literacy requirement

- In 2009, the Virginia Board of Education approved a graduation requirement for a stand-alone financial literacy course.
- The requirement applied to the graduating cohort of 2015, corresponding to students with birthdays after September 30, 1996.
- The course covers general economics and personal finance, including savings, banking transactions, credit-card terms, debt management, investment, taxes, and financial planning.
- The course is not merely embedded in mathematics or civics; students are required to attend a dedicated financial literacy class.

Interpretation: The policy is useful because the birth-date cutoff assigns otherwise similar students to different graduation cohorts. The treatment is plausibly unrelated to future criminal behavior except through exposure to financial literacy training.

2.2 Administrative criminal data

- The authors use administrative criminal court records from Virginia circuit and district courts for 2008–2020.

- Court records include defendant names, charges, hearing information, sentencing information, and dates of birth.
- These records are linked to residential addresses and demographic information from public-records data.
- The main outcome is whether an individual is charged with a financial crime.

Interpretation: The data are unusually granular: they observe individual criminal charges, not county-level crime rates or survey responses.

2.3 Financial-crime definition

The main outcome is

$$FinancialCrime_i = \mathbf{1}\{i \text{ is charged with embezzlement, fraud, forgery, or counterfeiting}\}.$$

The component outcomes are:

- **Embezzlement:** taking funds entrusted to the individual, typically in an employer-employee relationship.
- **Fraud:** general financial fraud, such as healthcare fraud, real-estate fraud, or identity theft.
- **Forgery:** falsifying financial documents or signatures.
- **Counterfeiting:** fraudulent use or creation of banknotes, coins, or bills.

Interpretation: Embezzlement is important because criminology research links it to a non-shareable financial problem: the individual faces financial pressure, has access to entrusted funds, and attempts to solve the problem privately.

2.4 Control variables and fixed effects

The baseline design uses high-dimensional fixed effects:

- Census tract $\times$ gender $\times$ race fixed effects;
- family-size fixed effects;
- homeownership fixed effects;
- running-variable controls around the birth-month threshold.

Interpretation: The preferred specification compares people in the same Census tract, with the same gender and race, and similar household characteristics, but different treatment status because of birth timing.

2.5 Mechanism datasets

The paper supplements court data with:

- FINRA National Financial Capability Study data to verify that the course is associated with higher financial literacy;
- Census/SIPP microdata to study financial behavior, such as credit-card debt, investments, and savings accounts;
- Suspicious Activity Reports and Virginia State Police incident data to test whether observed effects are due to lower detection rather than lower crime.

Interpretation: These auxiliary datasets move the paper beyond a reduced-form RDD. They test whether the mechanism is improved household balance sheets rather than reduced observability of crime.

3 Models and empirical specifications

3.1 Sharp regression discontinuity design

The core empirical specification is

$$Y_i = \alpha_0 + \alpha_1 \textit{FinancialLiteracy}_i + \alpha_2 \textit{MonthOfBirth}_i + \alpha_3 (\textit{FinancialLiteracy}_i \times \textit{MonthOfBirth}_i) + \alpha_4 X_i + \varepsilon_i. \quad (1)$$

- Y_i is a criminal-charge indicator, multiplied by 100 in the tables.
- $\textit{FinancialLiteracy}_i$ equals one if the individual is assigned to the cohort required to take financial literacy.
- $\textit{MonthOfBirth}_i$ is the running variable around the September 30, 1996 cutoff.
- X_i includes fixed effects and individual-level controls.
- The preferred bandwidth is 16 months on either side of the cutoff.

Interpretation: The coefficient α_1 is the discontinuity at the birth-date cutoff. Since outcomes are multiplied by 100, a coefficient of -0.259 means a decline of 0.259 percentage points.

3.2 Baseline outcome

For the main result,

$$Y_i = 100 \times \textit{FinancialCrime}_i.$$

Interpretation: The financial-crime rate is low in absolute terms, so percentage-point effects look small. Relative to the sample mean of about 0.708%, however, a 0.23–0.26 percentage-point drop is economically large.

3.3 Component-crime outcomes

For crime-type tests, the dependent variables are

$$Y_i \in \{100 \times \textit{Embezzlement}_i, 100 \times \textit{Fraud}_i, 100 \times \textit{Forgery}_i, 100 \times \textit{Counterfeit}_i\}.$$

Interpretation: This separates crimes likely driven by financial constraints from more sophisticated or opportunistic forms of financial misconduct.

3.4 Severity outcomes

The paper estimates the same RDD with:

$$Y_i \in \{100 \times \textit{MisdemeanorFinancialCrime}_i, 100 \times \textit{FelonyFinancialCrime}_i\}.$$

Interpretation: This asks whether the policy reduces only minor financial crime or also more socially costly felony financial crime.

3.5 Income heterogeneity

The mechanism specification includes an interaction:

$$Y_i = \beta_0 + \beta_1 \text{FinancialLiteracy}_i + \beta_2 \text{LowIncome}_i + \beta_3 (\text{FinancialLiteracy}_i \times \text{LowIncome}_i) + \Gamma X_i + u_i. \quad (2)$$

- LowIncome_i equals one if the individual lives in a Census block group in the lowest income quartile.
- β_3 measures whether the treatment effect is stronger for financially constrained individuals.

Interpretation: A negative β_3 supports the financial-constraints channel.

3.6 Indigent-defense outcome

Conditional on being charged with financial crime, the paper uses:

$$Y_i = 100 \times \text{Indigent}_i,$$

where $\text{Indigent}_i = 1$ if the defendant receives a public defender or self-represents.

Interpretation: Indigent defense is a legal-system proxy for financial distress among accused individuals. A lower indigent-defense rate among treated defendants suggests treated offenders are less financially distressed, consistent with fewer need-driven crimes.

3.7 Financial behavior outcomes

Using SIPP, the paper estimates RDDs for:

$$Y_i \in \{\text{CreditCardDTI}_i, \text{Investments}_i, \text{SavingsAccount}_i\}.$$

- CreditCardDTI is credit-card debt over income.
- Investments is stock and bond investments over total assets.
- SavingsAccount is an indicator for having a savings account.

Interpretation: These are direct balance-sheet mechanisms. Lower high-interest debt and higher saving/investment reduce the pressure that may induce embezzlement.

4 Identification and empirical design

4.1 What identifies the effect

- The cutoff is based on birth date, not on individual choice.
- Individuals near the cutoff are close in age and should face similar labor-market, school, and family environments.
- The policy was enacted after individuals were already assigned to class cohorts, limiting strategic selection into treatment.
- The main design uses local linear RDD with a 16-month bandwidth.

Interpretation: The identifying assumption is local comparability: people born just before and just after the cutoff would have followed similar crime paths absent the financial literacy requirement.

4.2 Why the design is credible

- The density of the running variable shows no evidence of manipulation around the cutoff.
- Observable characteristics do not jump around the cutoff.
- Placebo thresholds using adjacent cohorts show no discontinuity.
- The results are robust to alternative bandwidths, higher-order polynomials, and fixed-effect structures.

Interpretation: These checks are important because the treatment effect is identified from a discontinuity in birth-month assignment. If families manipulated birth timing or if other characteristics jumped at the cutoff, the design would be weaker.

4.3 Why this is not just a general education effect

- Individuals on either side of the cutoff are almost the same age.
- The placebo cohort tests do not show similar effects where no treatment difference exists.
- Treated individuals are not more likely to attend college or postgraduate programs in the mechanism tests.

Interpretation: The result is linked to the financial literacy course, not simply to more schooling or more time before entering the labor force.

4.4 Alternative mechanisms considered

The paper examines several alternatives:

- **Criminal sophistication:** treated individuals could become better at hiding crime.
- **Occupational selection:** treated individuals could avoid jobs with access to funds.
- **Detection:** reported crimes could fall even if actual crimes do not.
- **Mobility:** treated individuals could move out of Virginia.
- **Victimhood:** treated individuals could avoid becoming victims rather than avoid committing crimes.
- **School/family selection:** treatment effects could reflect differences in schools or families.

Interpretation: The evidence generally rejects these alternatives and points back to improved financial behavior and reduced financial constraints.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format. Adjacent figures and tables are grouped to save space and improve comparison. All visuals are directly cropped from the original paper.

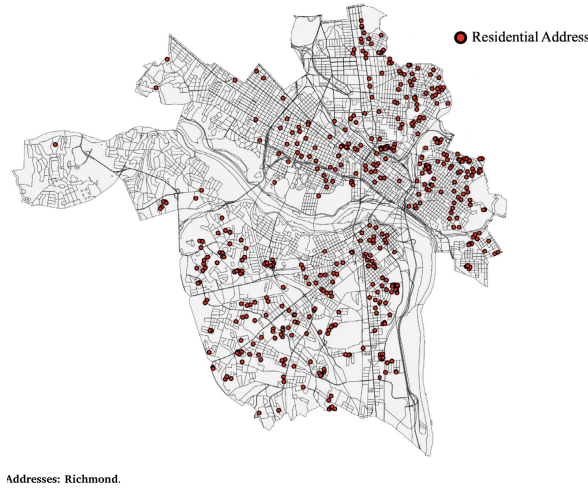
5.1 Figure 1 and Table 1: data granularity and summary statistics

Read:

- Figure 1 maps residential addresses of individuals charged with financial crimes in Richmond, Virginia.
- The map illustrates how granular the linked court-address data are.
- Table 1 shows that the mean financial-crime charge rate is 0.708%.

- The largest component categories are forgery at 0.255% and embezzlement at 0.243%, followed by fraud at 0.173% and counterfeiting at 0.078%.
- The sample is young, with mean age around 23.6, and 55.1% are treated by the financial literacy requirement.

Economic message: The data allow the authors to study ordinary individual financial crimes with much finer detail than county or state crime aggregates.



(a) Figure 1: residential addresses in Richmond.

Table 1
Summary statistics.
This table reports the summary statistics of the key variables used in the study. Crime data is from the district and circuit criminal court systems of the state of Virginia. Demographic variables are provided by L2, sourced from Virginia voter registration records. *Financial Literacy* is sourced from data on individuals' birth dates provided by L2. *Homeowner* is sourced from deeds data from county recorder offices in Virginia, while data on block group-level income (*Income (BG)*) is sourced from the US Census. We define variables in Section 3.

	Mean	SD	Median
Panel A: Crime Data (%)			
Financial Crime	0.708	8.383	0
Embezzlement	0.243	4.928	0
Fraud	0.173	4.155	0
Forgery	0.255	5.046	0
Counterfeit	0.078	2.797	0
Misdemeanor	0.296	5.430	0
Felony	0.407	6.367	0
Murder	0.022	1.494	0
Assault	0.511	7.133	0
Drugs	4.530	20.795	0
Larceny	2.161	14.539	0
Vandalism	0.008	0.900	0
Indigent	0.404	0.491	0
Panel B: Demographics			
Black or African American	0.171	0.377	0
White	0.542	0.498	1
Asian	0.043	0.202	0
Hispanic	0.085	0.279	0
Other	0.480	6.913	0
Male	0.446	0.497	0
Female	0.554	0.497	1
Family Size	2.426	1.211	2
Age	23.574	3.541	23
Panel C: Financial Literacy			
Financial Literacy	0.551	0.497	1
Panel D: Economic Variables			
Homeowner	0.072	0.259	0
Income (BG)	94,150.35	51,546.41	82,344

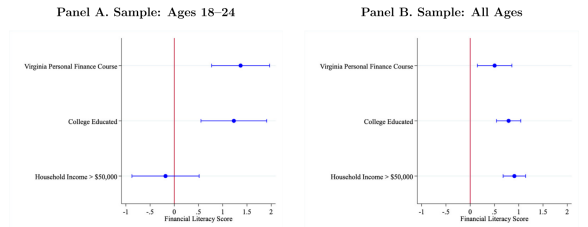
(b) Table 1: summary statistics.

5.2 Figure 2, Figure 3, and Table 2: first stage and baseline RDD

Read:

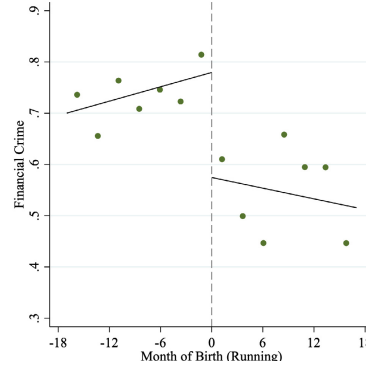
- Figure 2 shows that taking the Virginia personal finance course is strongly associated with higher tested financial literacy in FINRA survey data.
- For ages 18–24, the course is the strongest predictor of financial literacy.
- Figure 3 shows a downward discontinuity in financial crime at the treatment cutoff.
- Table 2 estimates treatment effects from -0.228 to -0.259 percentage points across specifications.
- The preferred specification with tract $\times$ gender $\times$ race, family size, and homeownership fixed effects gives -0.259 , or about a 36.6% drop relative to the mean.

Economic message: The course increases financial literacy, and the treated birth cohort commits fewer financial crimes.



Literacy: FINRA. Panel A: Sample: Ages 18-24. Panel B: Sample: All Ages. The figure shows the relative predictive power of taking the high school financial literacy course on tested financial literacy metrics for different samples. The metrics are Virginia Personal Finance Course, College Educated, and Household Income > \$50,000. The x-axis is Financial Literacy Score, ranging from -1 to 2. The y-axis lists the metrics. In Panel A, the scores are approximately: Virginia Personal Finance Course (1.5), College Educated (1.5), and Household Income > \$50,000 (0.5). In Panel B, the scores are approximately: Virginia Personal Finance Course (0.5), College Educated (0.5), and Household Income > \$50,000 (0.5).

(a) Figure 2: FINRA literacy evidence.



(b) Figure 3: RDD visual.

Table 2
Financial literacy and financial crime.

This table reports the effect of a financial literacy intervention on the probability of being charged with a financial crime. The dependent variable is Financial Crime, an indicator variable equal to 100 if an individual is charged with a financial crime such as financial fraud, embezzlement, forgery, or counterfeiting. The explanatory variable of interest is Financial Literacy, an indicator variable equal to one if an individual is required to take a financial literacy course to meet the requirements of graduating the Virginia high school system. The unit of observation is the individual. The bandwidth in the specification is 16 months on either side of the threshold, which we calculate as the optimal bandwidth. The linear controls include the running variable, Month of Birth, and Month of Birth $\times$ Financial Literacy. Standard errors are clustered at the level of the running variable. The statistics are denoted in parentheses. Statistical significance is indicated by ***, **, and * at the 1%, 5%, and 10% level, respectively. See Table 1 for sample descriptive characteristics.

	(1)	(2)	(3)	(4)
DV: Financial Crime				
Financial Literacy	-0.338*** (-2.788)	-0.237*** (-3.020)	-0.302*** (-3.488)	-0.309*** (-3.260)
Linear Controls	Yes	Yes	Yes	Yes
Trust-GenderRace FE	No	No	No	Yes
Trust-GenderRace FE	No	No	No	Yes
Trust FE	No	Yes	Yes	No
Gender FE	No	No	Yes	No
Race FE	No	No	Yes	No
Family Size FE	No	No	Yes	Yes
Homeownership FE	No	No	Yes	Yes
Observations	138,892	137,983	120,868	118,669

(c) Table 2: baseline estimates.

5.3 Table 3: robustness

Read:

- Panel A uses alternative bandwidths and polynomial orders.
- The financial-literacy coefficient remains negative across bandwidths from 6 to 24 months.
- Coefficients range from about -0.173 to -0.380 .
- Panel B uses alternative fixed-effect sets.
- The result remains negative and statistically significant with block-group, birth-month, gender, race, family-size, and homeownership fixed effects.

Economic message: The main result is not a fragile artifact of one bandwidth, one polynomial, or one fixed-effect structure.

Table 3**Alternative bandwidths, polynomials, and fixed effects.**

This table reports the effect of a financial literacy intervention on the probability of being charged with a financial crime using a variety of alternative specifications. Panel A shows results for various bandwidths and polynomials. Panel B shows results for alternative sets of fixed effects. The dependent variable of interest is *Financial Crime*, an indicator variable equal to 100 if an individual is charged with a financial crime such as financial fraud, embezzlement, forgery, or counterfeiting. The explanatory variable of interest is *Financial Literacy*, an indicator variable equal to one if an individual is required to take a financial literacy course to meet the requirements of graduating the Virginia high school system. The unit of observation is the individual. The bandwidth (*Bandwidth*) denotes the number of months on either side of the birth-month treatment assignment threshold. The linear controls include the running variable, *Month of Birth*, and *Month of Birth* $\times$ *Financial Literacy*. The 2nd and 3rd degree controls indicate the associate variables for 2nd and 3rd degree polynomials. Standard errors are clustered at the level of the running variable. The *t*-statistics are denoted in parenthesis. Statistical significance is indicated by ***, **, and * at the 1%, 5%, and 10% level, respectively. See Table 1 for sample descriptive characteristics.

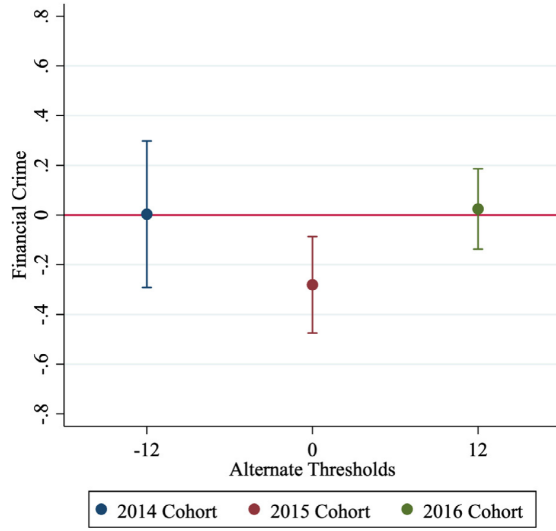
Panel A: Alternate Bandwidths and Polynomials						
	(1)	(2)	(3)	(4)	(5)	(6)
DV: Financial Crime						
Financial Literacy	-0.380** (-2.965)	-0.281*** (-3.005)	-0.217*** (-2.763)	-0.173** (-2.378)	-0.298*** (-3.207)	-0.276** (-2.398)
Linear Controls	Yes	Yes	Yes	Yes	Yes	Yes
2nd Degree Controls	No	No	No	No	Yes	Yes
3rd Degree Controls	No	No	No	No	No	Yes
TractxGenderxRace FE	Yes	Yes	Yes	Yes	Yes	Yes
Family Size FE	Yes	Yes	Yes	Yes	Yes	Yes
Homeowner FE	Yes	Yes	Yes	Yes	Yes	Yes
Bandwidth	6	12	18	24	24	24
Observations	38 304	82 690	126 026	168 112	168 112	169 994
R-squared	0.162	0.107	0.079	0.066	0.066	0.023
Panel B: Alternate Fixed Effects						
	(1)	(2)	(3)	(4)	(5)	(6)
DV: Financial Crime						
Financial Literacy	-0.209** (-2.100)	-0.235** (-2.201)	-0.361** (-2.693)	-0.229*** (-4.002)	-0.247*** (-4.500)	-0.245*** (-4.461)
Linear Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block GroupxGenderxRace FE	No	No	Yes	No	No	No
Birth MonthxGenderxRace FE	No	No	No	No	No	Yes
Block Group FE	Yes	Yes	No	No	No	No
Birth Month FE	No	Yes	No	Yes	Yes	No
Gender FE	No	Yes	No	No	Yes	No
Race FE	No	Yes	No	No	Yes	No
Family Size FE	No	Yes	Yes	No	Yes	Yes
Homeowner FE	No	Yes	Yes	No	Yes	Yes
Observations	115 139	99 475	65 755	138 892	122 486	122 486
R-squared	0.235	0.243	0.344	0.000	0.005	0.006

Figure 3: Table 3: alternative bandwidths, polynomials, and fixed effects.**5.4 Figure 4 and Table 4: placebo thresholds and crime types**

Read:

- Figure 4 tests adjacent cohort thresholds.
- Only the 2015 treated-cohort threshold shows a statistically significant discontinuity.
- The 2014 and 2016 placebo thresholds are close to zero.
- Table 4 shows that the treatment effect is concentrated in embezzlement.
- The embezzlement coefficient is -0.143 percentage points and statistically significant.
- Fraud, forgery, and counterfeiting estimates are negative but not statistically significant.

Economic message: The placebo result strengthens the RDD design. The crime-type result supports the financial-constraints mechanism, because embezzlement is the crime most associated with financial pressure.



(a) Figure 4: alternative thresholds.

Table 4

Types of financial crimes.

This table reports the effect of a financial literacy intervention on the probability of being charged with various types of financial crimes. The dependent variables of interest are *Embezzlement*, *Fraud*, *Forgery*, and *Counterfeit*, which correspond to indicator variables equal to 100 if the individual has been charged with the respective financial crime. The explanatory variable of interest is *Financial Literacy*, an indicator variable equal to one if an individual is required to take a financial literacy course to meet the requirements of graduating the Virginia high school system. The unit of observation is the individual. The bandwidth in the specification is 16 months on either side of the threshold, which we calculate as the optimal bandwidth. The linear controls include the running variable, *Month of Birth*, and *Month of Birth* $\times$ *Financial Literacy*. Standard errors are clustered at the level of the running variable. The *t*-statistics are denoted in parenthesis. Statistical significance is indicated by ***, **, and * at the 1%, 5%, and 10% level, respectively. See Table 1 for sample descriptive characteristics.

	(1)	(2)	(3)	(4)
	Embezzlement	Fraud	Forgery	Counterfeit
Financial Literacy	-0.143** (-2.461)	-0.042 (-0.819)	-0.028 (-0.692)	-0.036 (-1.203)
Linear Controls	Yes	Yes	Yes	Yes
Tract $\times$ Gender $\times$ Race FE	Yes	Yes	Yes	Yes
Family Size FE	Yes	Yes	Yes	Yes
Homeowner FE	Yes	Yes	Yes	Yes
Observations	118 689	118 689	118 689	118 689
R-squared	0.081	0.075	0.074	0.076

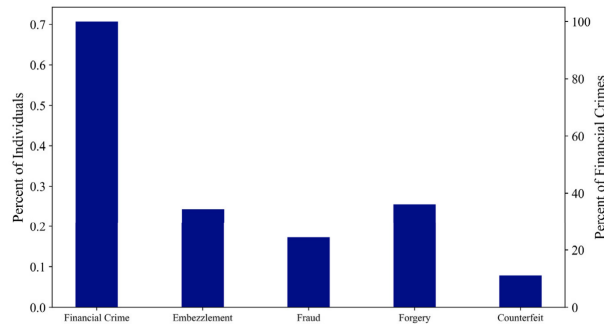
(b) Table 4: types of financial crimes.

5.5 Figure 5 and Figure 6: financial-crime composition and type-specific RDDs

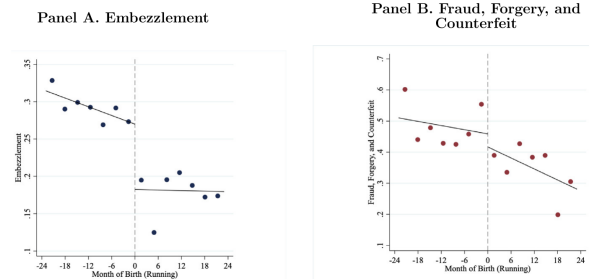
Read:

- Figure 5 shows that financial crime is rare overall, but embezzlement and forgery are the most common components.
- Embezzlement accounts for a large share of financial crimes in the sample.
- Figure 6 shows the visual discontinuity by crime type.
- The discontinuity is sharp for embezzlement.
- Fraud, forgery, and counterfeiting show no comparable discontinuity.

Economic message: The visual evidence reinforces the table result: financial literacy reduces the type of financial crime most plausibly tied to personal balance-sheet stress.



(a) Figure 5: composition of financial crime.



(b) Figure 6: RDD by crime type.

5.6 Table 5 and Table 6: severity and non-financial crime

Read:

- Table 5 shows that financial literacy reduces both misdemeanor and felony financial crime.
- The misdemeanor coefficients are about -0.098 percentage points.
- The felony coefficients are about -0.139 to -0.140 percentage points.
- Table 6 shows no meaningful treatment effects for murder, assault, drugs, vandalism, or larceny.

Economic message: The course reduces financial crime across severity levels, but it does not create a general reduction in all crime. The effect is specific to financially motivated behavior.

Table 5

Crime severity.

This table reports the effect of a financial literacy intervention on the probability of being charged with a misdemeanor or felony financial crime. The dependent variable in columns 1 and 2 is *Misdemeanor*, which is defined as an indicator variable equal to 100 if an individual is charged with a misdemeanor financial crime. The dependent variable in columns 3 and 4 is *Felony*, which is an indicator variable equal to 100 if an individual is charged with a felony financial crime. The explanatory variable of interest is *Financial Literacy*, an indicator variable equal to one if an individual is required to take a financial literacy course to meet the requirements of graduating the Virginia high school system. The unit of observation is the individual. The bandwidth in the specification is 16 months on either side of the threshold, which we calculate as the optimal bandwidth. The linear controls include the running variable, *Month of Birth*, and *Month of Birth* $\times$ *Financial Literacy*. Standard errors are clustered at the level of the running variable. The *t*-statistics are denoted in parenthesis. Statistical significance is indicated by ***, **, and * at the 1%, 5%, and 10% level, respectively. See Table 1 for sample descriptive characteristics

	(1)	(2)	(3)	(4)
	Misdemeanor	Misdemeanor	Felony	Felony
Financial Literacy	-0.098^{**} (-2.464)	-0.098^{**} (-2.488)	-0.139^{**} (-2.076)	-0.140^{**} (-2.075)
Linear Controls	Yes	Yes	Yes	Yes
Tract $\times$ Gender $\times$ Race FE	Yes	Yes	Yes	Yes
Family Size FE	No	Yes	No	Yes
Homeowner FE	No	Yes	No	Yes
Observations	118 689	118 689	118 689	118 689
R-squared	0.074	0.074	0.085	0.086

(a) Table 5: crime severity.

Table 6

Other non-financial crimes.

This table reports the effect of a financial literacy intervention on the probability of being charged with a crime unrelated to finance. The dependent variables of interest are *Murder*, *Assault*, *Drugs*, *Vandalism*, and *Larceny*, which correspond to indicator variables equal to 100 if the individual has been charged with the respective crime. The explanatory variable of interest is *Financial Literacy*, an indicator variable equal to one if an individual is required to take a financial literacy course to meet the requirements of graduating the Virginia high school system. The unit of observation is the individual. The bandwidth in the specification is 16 months on either side of the threshold, which we calculate as the optimal bandwidth. The linear controls include the running variable, *Month of Birth*, and *Month of Birth* $\times$ *Financial Literacy*. Standard errors are clustered at the level of the running variable. The *t*-statistics are denoted in parenthesis. Statistical significance is indicated by ***, **, and * at the 1%, 5%, and 10% level, respectively. See Table 1 for sample descriptive characteristics.

	(1)	(2)	(3)	(4)	(5)
	Murder	Assault	Drugs	Vandalism	Larceny
Financial Literacy	-0.000 (-0.025)	0.071 (1.220)	-0.112 (-0.321)	0.002 (0.227)	-0.036 (-0.196)
Linear Controls	Yes	Yes	Yes	Yes	Yes
Tract $\times$ Gender $\times$ Race FE	Yes	Yes	Yes	Yes	Yes
Family Size FE	Yes	Yes	Yes	Yes	Yes
Homeowner FE	Yes	Yes	Yes	Yes	Yes
Observations	118 689	118 689	118 689	118 689	118 689
R-squared	0.070	0.074	0.108	0.030	0.092

(b) Table 6: other non-financial crimes.

5.7 Figure 7 and Table 7: income and financial constraints

Read:

- Figure 7 maps Virginia income quartiles and shows a monotonic relation between low local income and financial-crime risk.
- Individuals in the lowest-income block groups have a much higher probability of being charged with financial crime.
- Table 7 interacts treatment with low-income location.
- The interaction coefficients are negative, around -0.295 to -0.343 percentage points.
- The low-income main effect is positive, consistent with financial crime being more likely in constrained areas.

Economic message: Financial literacy matters more where financial constraints are likely stronger, which supports the paper’s main mechanism.

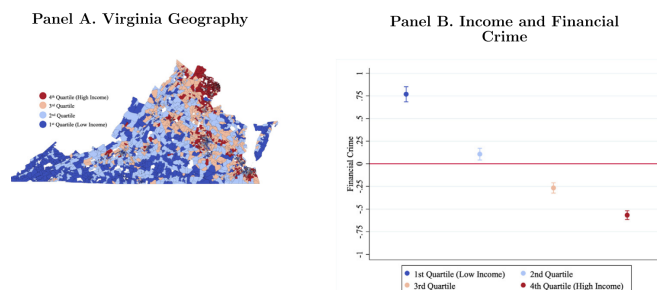


Figure 7: income and financial constraints. Panel A presents a map of the state of Virginia showing income quartiles. The lowest income quartile is highlighted in blue, and the highest income quartile is highlighted in red. Panel B presents coefficient estimates relating to local income and financial crimes in Virginia.

(a) Figure 7: income and financial constraints.

Table 7

Heterogeneity by income level.

This table reports the effect of a financial literacy intervention by income level on the probability of being charged with a financial crime. The dependent variable is *Financial Crime*, an indicator variable equal to 100 if an individual is charged with a financial crime such as financial fraud, embezzlement, forgery, or counterfeiting. *Financial Literacy* is an indicator variable equal to one if an individual is required to take a financial literacy course to meet the requirements of graduating the Virginia high school system. *Low Income* is an indicator equal to one if an individual lives in a Census block group in the lowest quartile of income in Virginia. The unit of observation is the individual. The bandwidth in the specification is 16 months on either side of the threshold, which we calculate as the optimal bandwidth. The linear controls include the running variable, *Month of Birth*, and *Month of Birth* $\times$ *Financial Literacy*. Standard errors are clustered at the level of the running variable. The *t*-statistics are denoted in parenthesis. Statistical significance is indicated by ***, **, and * at the 1%, 5%, and 10% level, respectively. See Table 1 for sample descriptive characteristics.

	(1)	(2)	(3)	(4)
DV: Financial Crime				
Financial Literacy $\times$ Low Income	-0.300** (-2.276)	-0.295** (-2.236)	-0.343** (-2.419)	-0.307* (-1.799)
Low Income	0.814*** (7.525)	0.570*** (3.937)	0.480*** (3.278)	0.373** (2.278)
Financial Literacy	-0.070 (-0.910)	-0.067 (-0.864)	-0.048 (-0.618)	-0.048 (-0.494)
Linear Controls	Yes	Yes	Yes	Yes
Tract $\times$ Gender $\times$ Race FE	No	No	No	Yes
Tract FE	No	Yes	Yes	No
Gender FE	No	No	Yes	No
Race FE	No	No	Yes	No
Family Size FE	No	No	Yes	Yes
Homeowner FE	No	No	Yes	Yes
Observations	99,276	99,275	87,632	85,589
R-squared	0.001	0.017	0.021	0.095

(b) Table 7: low-income heterogeneity.

5.8 Table 8, Figure 8, and Table 9: financial distress and financial behavior

Read:

- Table 8 shows that treated individuals charged with financial crime are 19–24 percentage points less likely to use indigent defense.
- Figure 8 shows discontinuities in high-interest credit, investments, and savings accounts.
- Table 9 shows that treatment reduces credit-card debt-to-income by about 0.029–0.030.
- Treatment increases investments by about 0.138.
- Treatment increases savings-account ownership by about 0.322–0.328.

Economic message: This is the strongest mechanism evidence. Financial literacy changes balance sheets in the direction that should reduce the incentive to commit need-driven financial crime.

Table 8

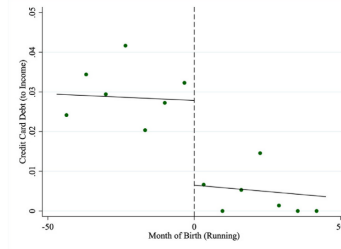
Indigent defense.

This table reports the effect of a financial literacy intervention on the probability of employing an indigent defense. The dependent variable is *Indigent*, which is defined as an indicator variable equal to 100 if an individual accused of financial crime is assigned a public defender or self-represents in district court. The explanatory variable of interest is *Financial Literacy*, an indicator variable equal to one if an individual is required to take a financial literacy course to meet the requirements of graduating the Virginia high school system. The unit of observation is the individual conditional on being charged with a financial crime in district court. The bandwidth in the specification is 16 months on either side of the threshold, which we calculate as the optimal bandwidth. The linear controls include the running variable, *Month of Birth*, and *Month of Birth* $\times$ *Financial Literacy*. Standard errors are clustered at the level of the running variable. The *t*-statistics are denoted in parenthesis. Statistical significance is indicated by ***, **, and * at the 1%, 5%, and 10% level, respectively. See Table 1 for sample descriptive characteristics.

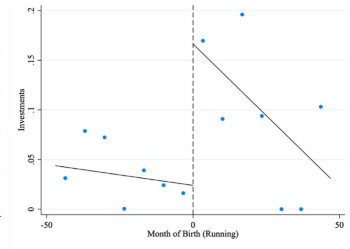
	(1)	(2)	(3)	(4)
	Indigent	Indigent	Indigent	Indigent
Financial Literacy	-0.212** (-2.603)	-0.238** (-3.331)	-0.192* (-1.912)	-0.209* (-1.985)
Linear Controls	Yes	Yes	Yes	Yes
Tract $\times$ Gender $\times$ Race FE	No	No	Yes	Yes
Tract FE	Yes	Yes	No	No
Gender FE	No	Yes	No	No
Race FE	No	Yes	No	No
Family Size FE	No	Yes	No	Yes
Homeowner FE	No	Yes	No	Yes
Observations	653	589	266	266

(a) Table 8: indigent defense.

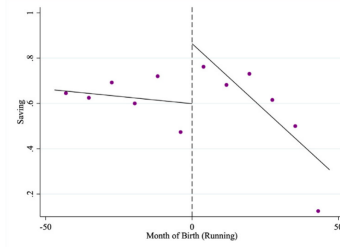
Panel A. High Interest Credit



Panel B. Investments



Panel C. Savings Accounts



(b) Figure 8: financial behavior discontinuities.

Table 9

Financial decision-making.

This table reports the effect of a financial literacy intervention on financial decision-making. The dependent variable in columns 1 and 2 is *Credit Card DTI*, which is defined as the ratio of credit card debt to income. The dependent variable in columns 3 and 4 is *Investments*, which is defined as the ratio of stock and bond investments to total assets. The dependent variable in columns 5 and 6 is *Savings Account*, which is defined as a dummy variable equal to one if an individual maintains a savings account. The explanatory variable of interest is *Financial Literacy*, an indicator variable equal to one if an individual is required to take a financial literacy course to meet the requirements of graduating the Virginia high school system. The unit of observation is the individual-year. The bandwidth in the specification is 48 months on either side of the threshold. The linear controls include the running variable, *Month of Birth*, and *Month of Birth* $\times$ *Financial Literacy*. Standard errors are clustered at the level of the running variable. The *t*-statistics are denoted in parenthesis. Statistical significance is indicated by ***, **, and * at the 1%, 5%, and 10% level, respectively. See Table 1 for sample descriptive characteristics.

	(1)	(2)	(3)	(4)	(5)	(6)
	Credit Card DTI	Credit Card DTI	Investments	Investments	Savings Account	Savings Account
Financial Literacy	-0.029* (-1.691)	-0.030* (-1.719)	0.138** (2.233)	0.138** (2.293)	0.322** (2.292)	0.328** (2.354)
Linear Controls	Yes	Yes	Yes	Yes	Yes	Yes
Income and Wealth	No	Yes	No	Yes	No	Yes
Gender	Yes	Yes	Yes	Yes	Yes	Yes
Race	Yes	Yes	Yes	Yes	Yes	Yes
US Citizen	Yes	Yes	Yes	Yes	Yes	Yes
Family Size	Yes	Yes	Yes	Yes	Yes	Yes
Employed	Yes	Yes	Yes	Yes	Yes	Yes
Education	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Observations	176	176	208	208	214	214
R-squared	0.273	0.283	0.169	0.211	0.212	0.219

Figure 9: Table 9: financial decision-making.

6 Short summary

- The paper studies whether financial literacy can reduce financial crime.
- The setting is Virginia's mandatory high-school financial literacy requirement beginning with the 2015 graduating cohort.
- The identification strategy is a sharp RDD around the September 30, 1996 birth-date cutoff.
- The data link Virginia court records to address and demographic information for a near-complete population sample.
- The preferred baseline coefficient is about -0.259 percentage points.

- Since the mean financial-crime rate is about 0.708%, this implies a roughly 37% reduction.
- The treatment effect is concentrated in embezzlement.
- The treatment also reduces both misdemeanor and felony financial crime.
- The treatment does not reduce unrelated crimes such as murder, assault, drugs, vandalism, or larceny.
- Low-income neighborhoods have higher financial-crime risk, and the treatment effect is stronger in these areas.
- Treated financial-crime defendants are less likely to use indigent defense.
- SIPP evidence shows lower credit-card debt, higher investments, and higher savings-account ownership after treatment.
- The paper rejects several alternative explanations, including criminal sophistication, occupational selection, detection, migration, victimhood, and school selection.
- The central mechanism is improved household financial resilience.

7 Conclusion

7.1 Core conclusion

Freed and Hackney show that financial literacy training can reduce financial crime. A mandated high-school financial literacy course lowers the probability of being charged with financial crime among young adults in Virginia, with the effect concentrated in embezzlement.

7.2 Economic intuition

Financial crime often involves monetary motivation and abuse of trust. If financial literacy improves debt management, saving, and investment behavior, then individuals are less likely to face the severe non-shareable financial problems that make embezzlement attractive. The course does not appear to make people generally less criminal; it specifically reduces financially motivated crime.

7.3 Takeaway

The paper expands the welfare case for financial education. The benefits are not limited to better household portfolios or lower debt. Financial literacy can generate external social benefits by reducing financial misconduct that harms employers, customers, firms, and the legal system.